Advanced Biotechnology

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CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 15

Cell free systems

+

Bottom-up and Top-down synthetic biology (Synthetic Cells)



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Lecture outline

1) Cell free systems and bioprocessing
   - Example: mRNA production
2) Top-down synthetic biology: Minimal cells
3) Bottom-up synthetic biology: Synthetic cells

Learning outcomes

1) Discuss basic principles and components in cell free systems
2) Discuss advantages and disadvantages of cell-free vs cell-based bioprocessing
3) Define top-down and bottom-up synthetic biology
4) Discuss the ideas, construction and interest around minimal cells
5) Discuss the philosophy, common features and applications of synthetic cells

Lecture outline

1) **Cell free systems and bioprocessing**
   - **Example: mRNA production**
2) Top-down synthetic biology: Minimal cells
3) Bottom-up synthetic biology: Synthetic cells

Challenges with cell-based biotechnology

The **use of live biological cells poses numerous challenges** to bioprocessing

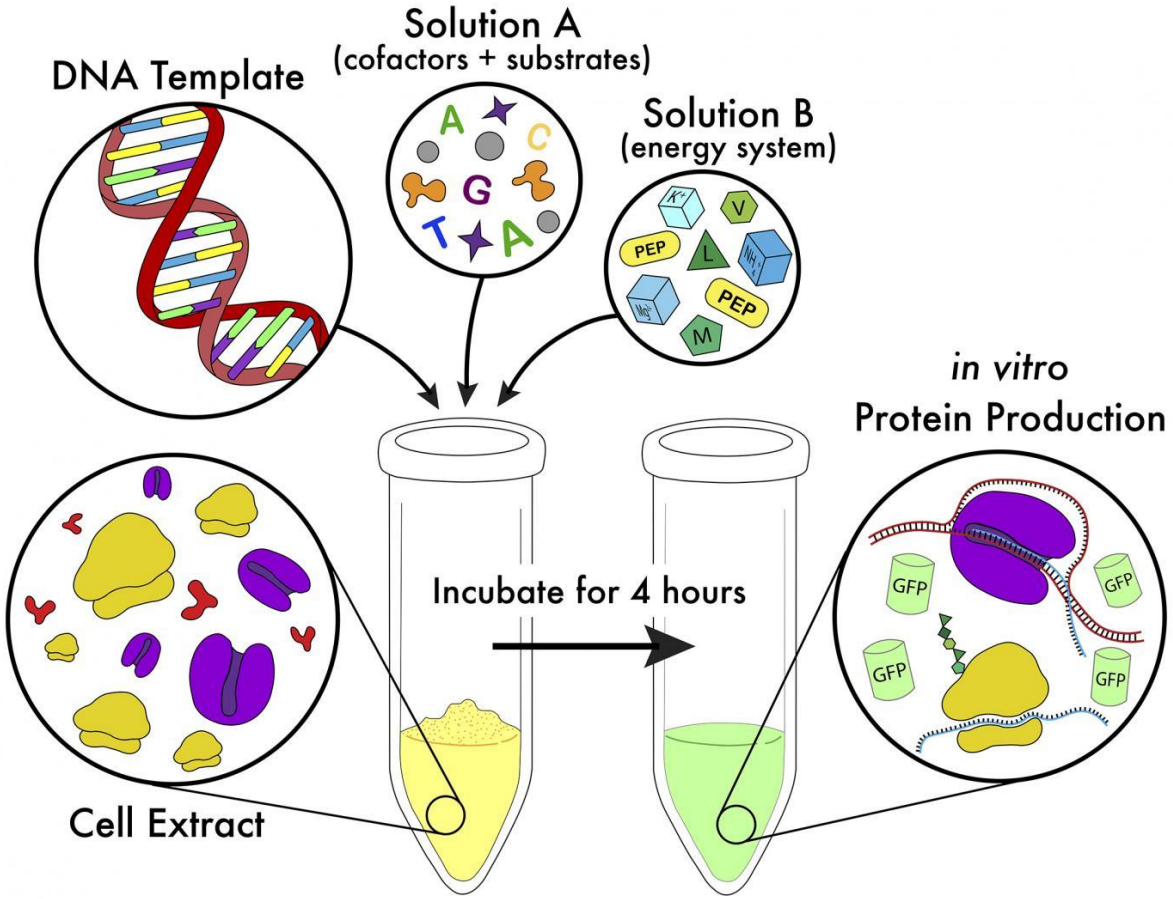
- Need to develop **genetically engineering cells**, which carries an economic and regulatory burden (GMO regulations)
- Some cell lines may be pathogenic, adding to safety concerns
- Some compounds produced may be **toxic to the cells**, limiting the quantities that can be expressed

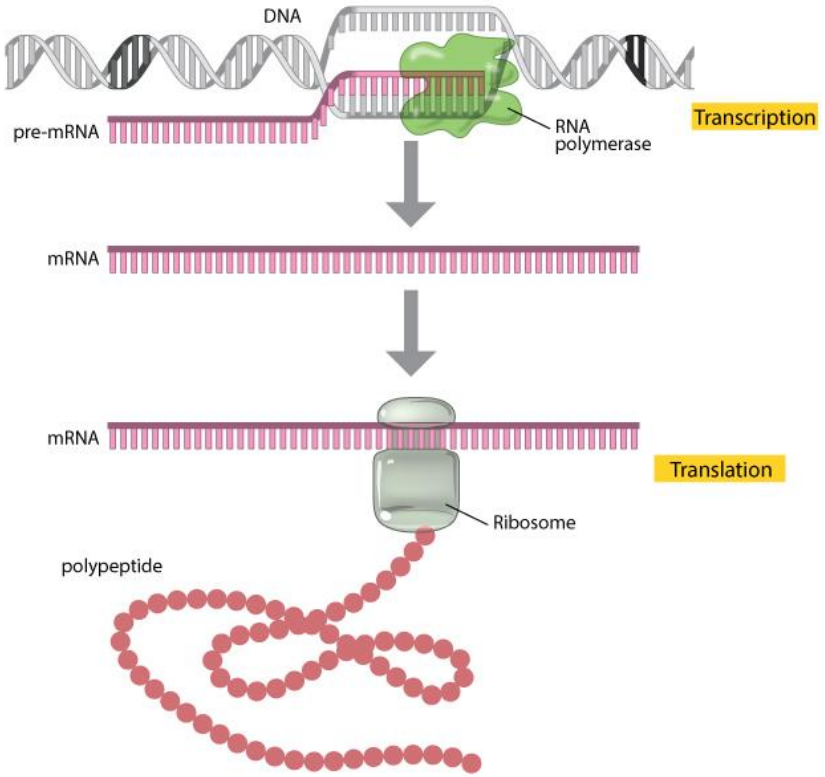
Challenges with cell-based biotechnology

- In **upstream processing**, challenges associated to maintaining the cells in optimal conditions (oxygenation, nutrients, limit shearing, avoid sedimentation)
- In **scaling up**, challenges associated to predicting biological response (slow optimisation cycles)
- In **downstream processing**, need for energy intensive cell disruption and removal steps

Cell free bioprocessing

Cell free bioprocessing uses cell lysates or cell extracts to sustain transcription and translation in vitro





Cell free TXTL systems

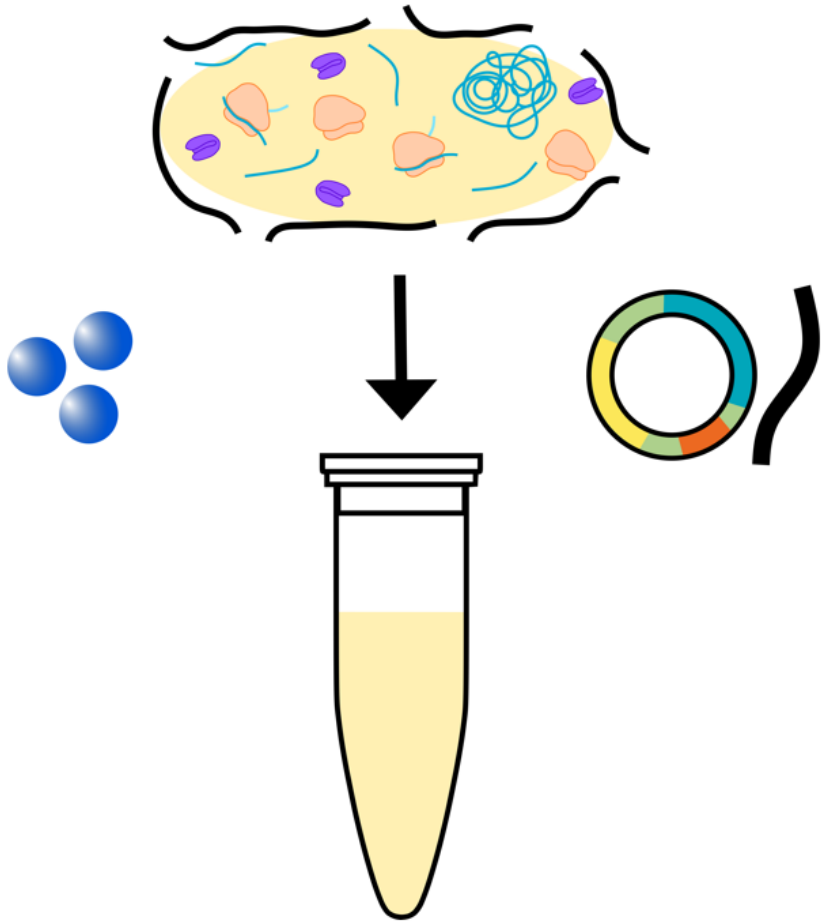
Cell free Transcription (TX) Translation (TL) systems come in two forms:

1) **"Crude" cell extracts**

1) **PURE system**

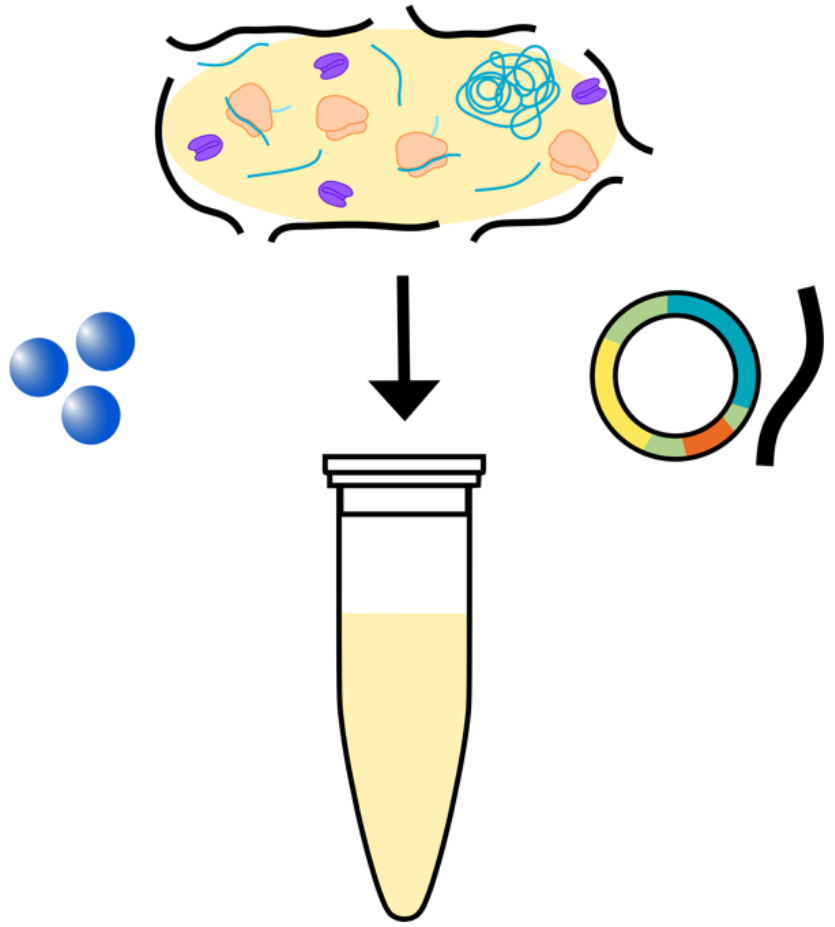
Cell free TXTL systems: Cell extracts

- Cell lysates prepared by **lysing cells** to extract transcription and translation machinery (RNA polymerases, ribosomes, tRNA)
- Can be prepared from **different cell types**, depending on the application, including bacteria, archea, yeast, plant (wheat germ) and animal cell lines (rabbit retyculocites)
- Lysates need to be **clarified** by removing bits of membrane/cell wall (as we studied) and cellular DNA (we want to put in our own!), producing a clarified **cell extract**



Cell free TXTL systems: Cell extracts

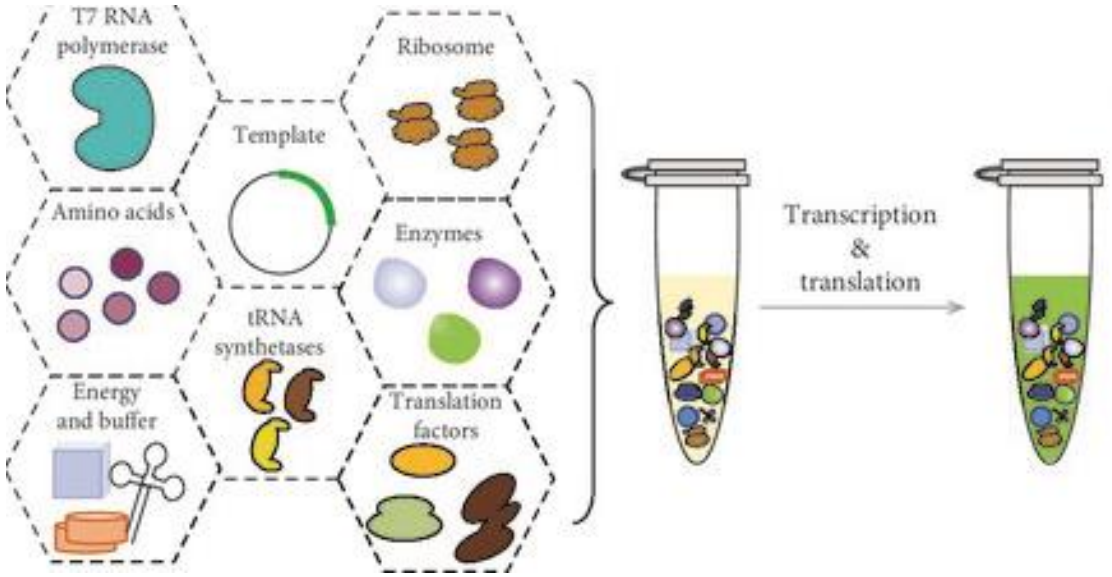
- Cell extracts are complemented with **building blocks** for making mRNA and proteins (NTPs, amino-acids), **crowding agents**, **energy substrates** and metabolic enzymes used to replenish ATP
- The blueprint for the protein we would like to produce is then added in the form of a **linear or circular** (plasmid) **DNA template**



Credit: Eugenia Romantseva (NIST)

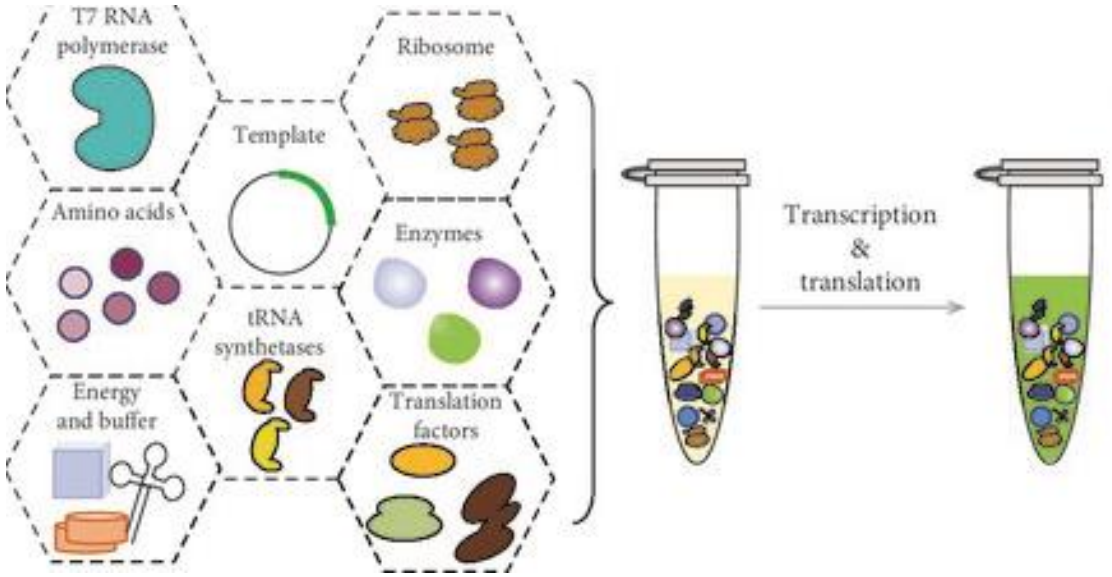
Cell free TXTL systems: PURE System

- **Protein synthesis Using Purified recombinant Elements** (PURE)
- Mixture of **36 recombinant proteins** expressed and purified + ribosomes, tRNAs, and all necessary building blocks and small molecules needed for protein expression



Cell free TXTL systems: PURE System

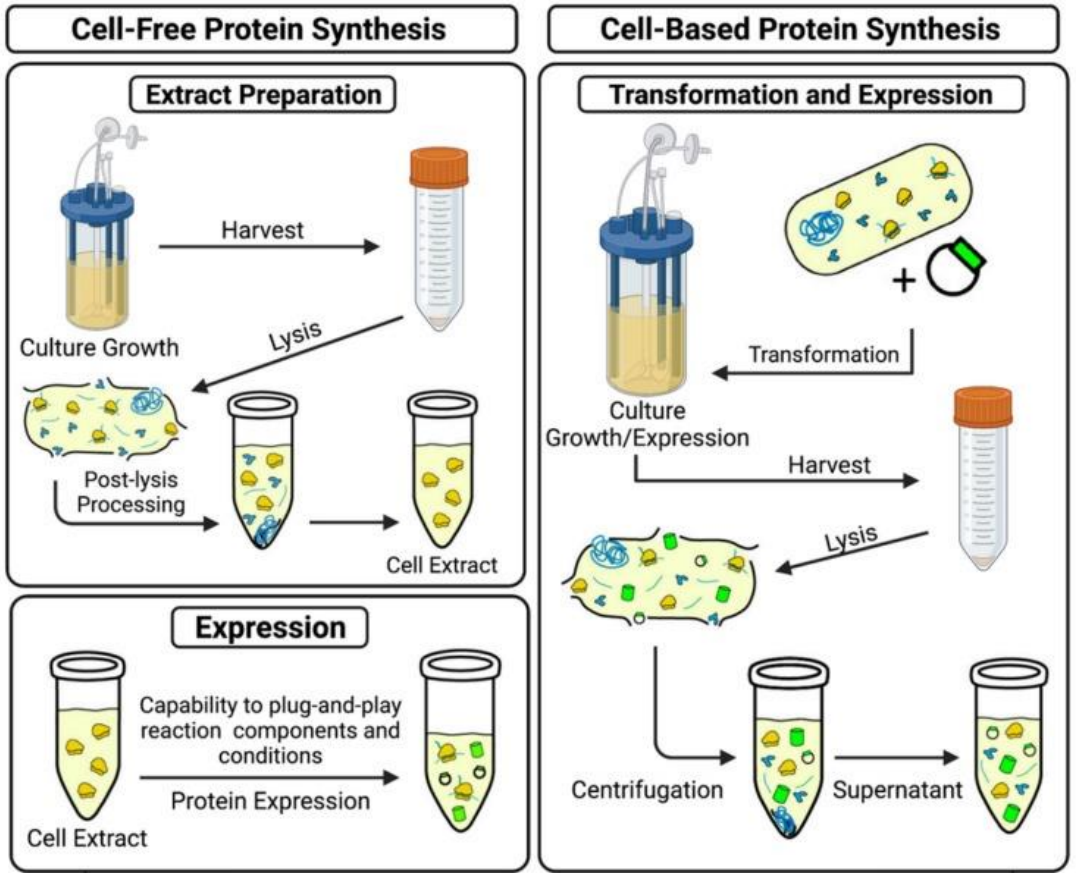
- Advantages over cell extracts
  - Allows precise control over all components
  - Can be used with non-natural amino-acids more easily
- Disadvantage over cell extracts
  - More expensive
  - Generally, less efficient (lower yield for protein production)



Cell-Free vs Cell-Based

Advantages of Cell Free

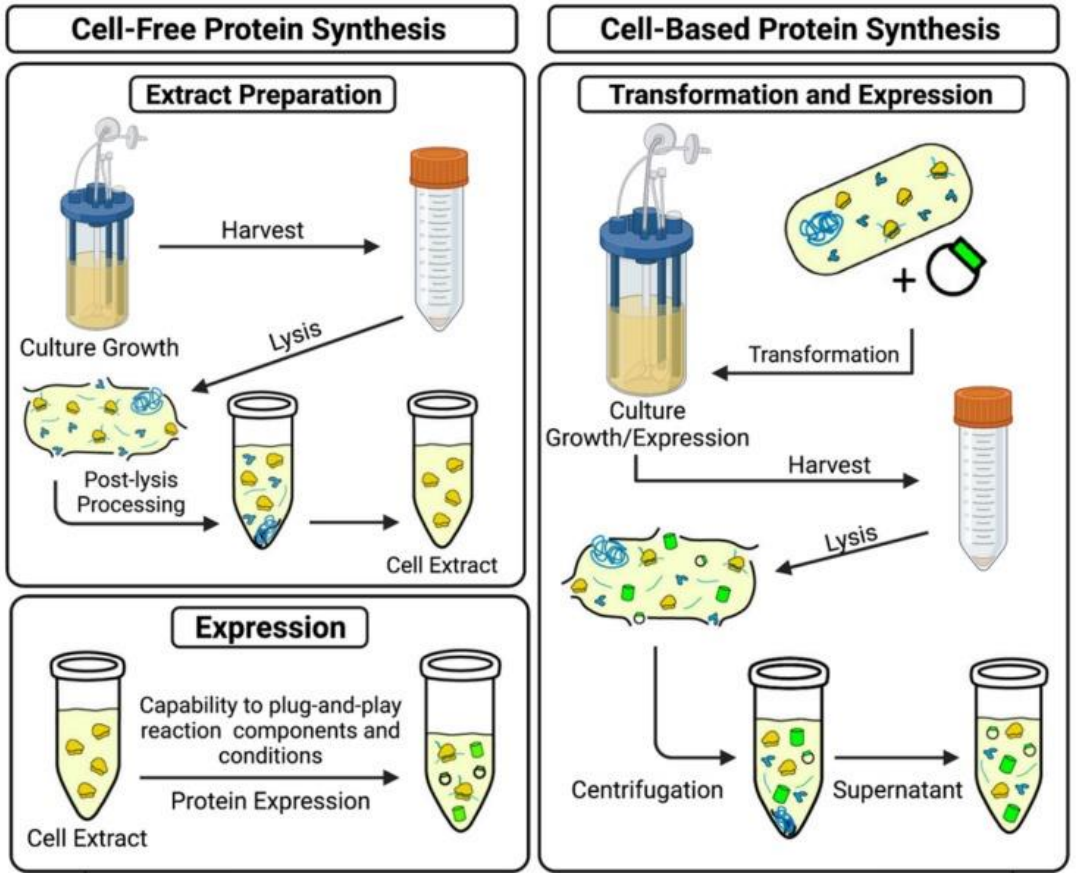
- Cell extract (or PURE), once produced, constitutes a **universal expression platform**—we just to add the gene that needs that needs expressing
- **No new GMO produced** (besides those needed to make the cell extract/PURE)as DNA templates are only added to the cell-free reaction mix. No regulatory or safety concerns



Cell-Free vs Cell-Based

… more advantages of Cell Free

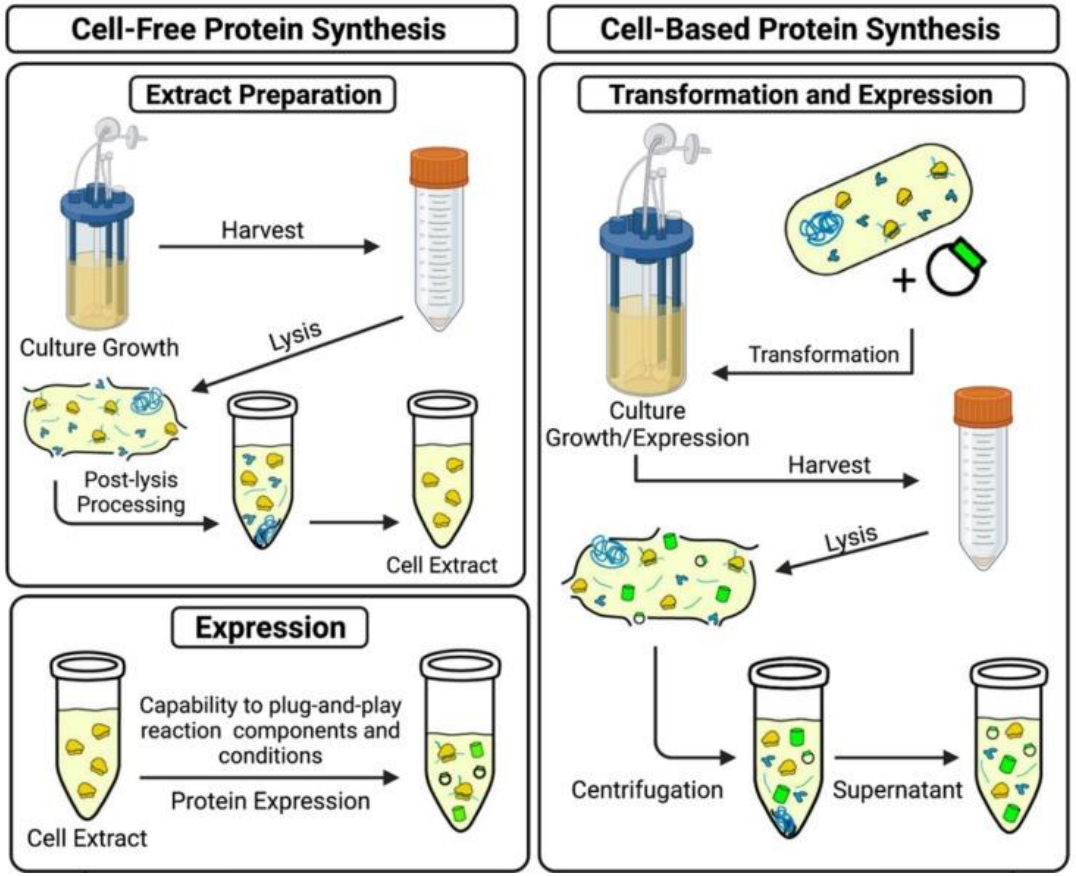
- Easier to produce compounds that would be **toxic for cells**
- **Mitigates challenges associated with fermentation** (oxygenation, mechanical perturbation…)
- Easier downstream processing: **No need for cell removal or cell disruption**



Cell-Free vs Cell-Based

… more advantages of Cell Free

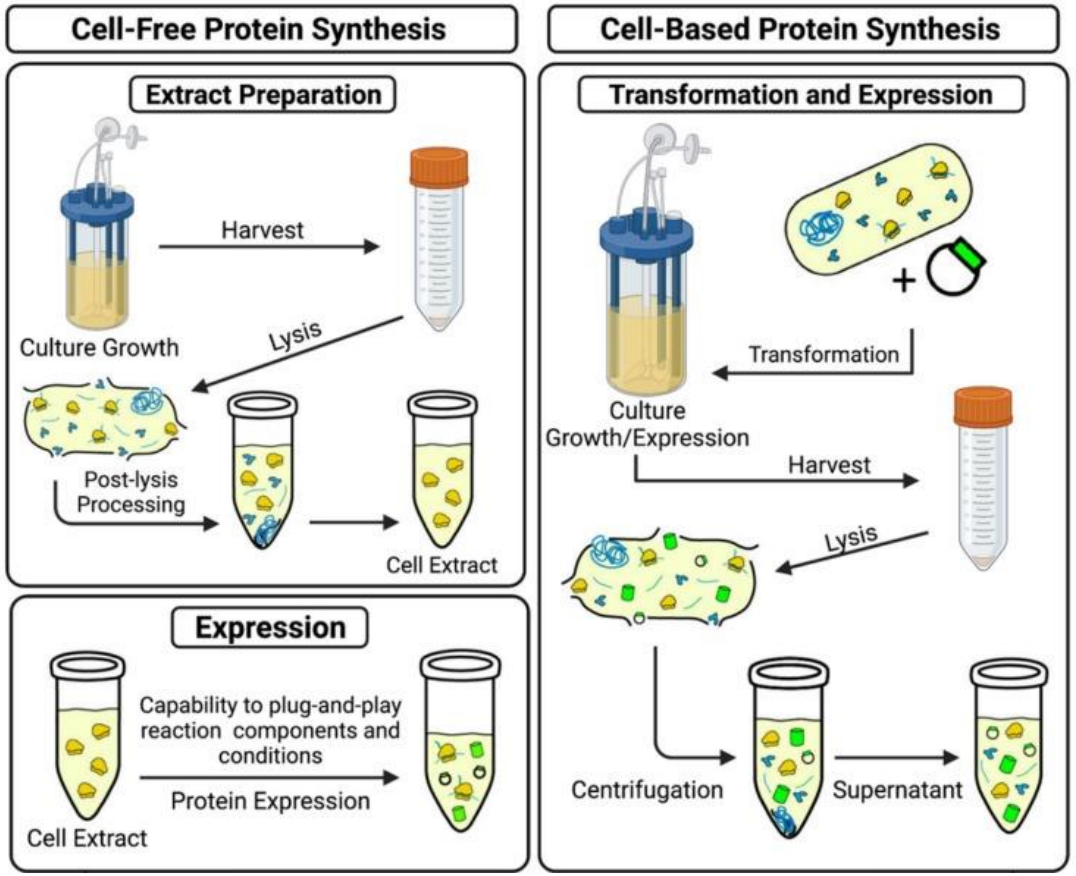
- **Faster optimisation cycles**
- **Easier to automate** using liquid handling robots, micro/millifluidics
- Cell extracts can be be **lyophilised** for easy and cheap transport
- Can function with **linear DNA** (rather than plasmids)
- Easier to **monitor reactions**
- More **controlled environment**



Cell-Free vs Cell-Based

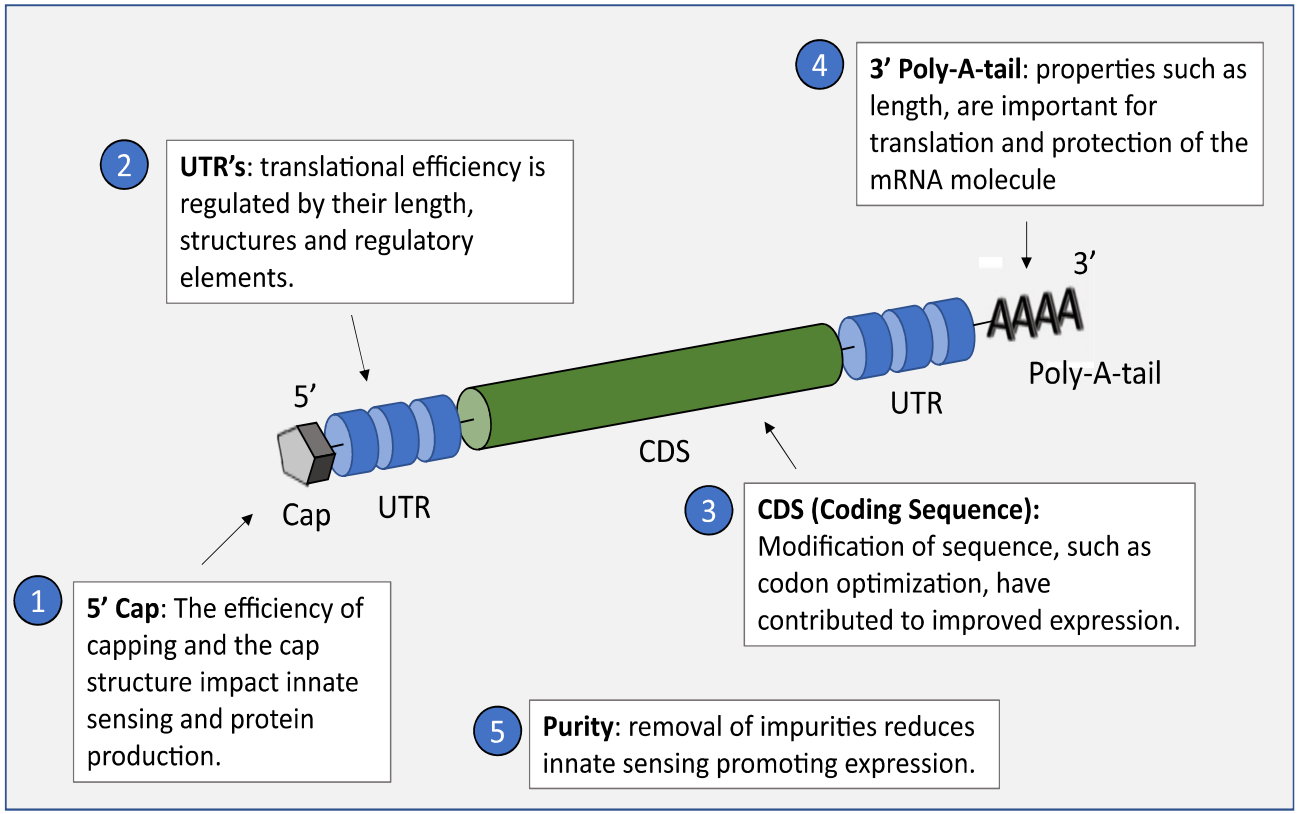
Disadvantages of Cell Free

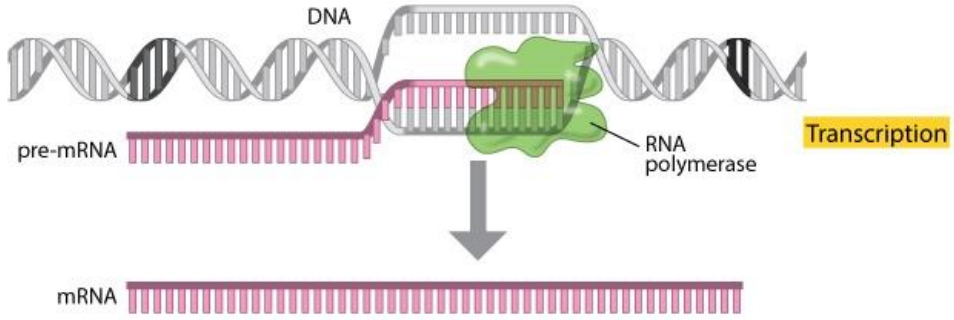
- Yields are **generally lower** compared to a cell-based process (adds to cost)
- **Cell free systems do not self-replicate** (adds to cost)
- **Not all products** can be effectively produced cell free. E.g. membrane proteins are challenging



Example: production of mRNA for vaccines

- mRNA used for COVID vaccines (e.g. Pfizer and Moderna) are made using a cell free process, namely **in vitro transcription** (IVT)





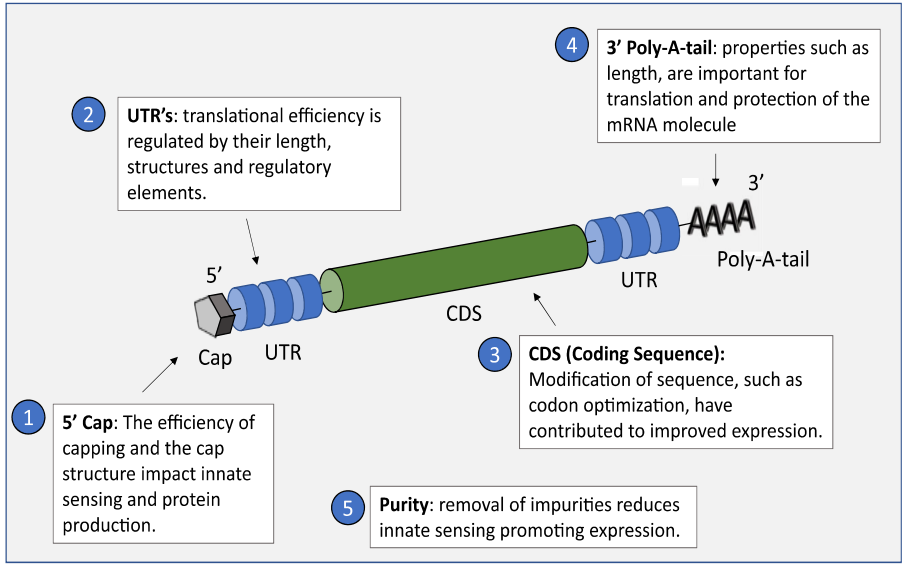
Example: production of mRNA for vaccines

IVT is carried out in mixtures containing

- **RNA polymerase**. A common choice is T7 RNA polymerase from the T7 bacteriophage. It has very high transcription rate, specificity for its promoter region, and low error rate
- A **DNA template** with a sequence of bases (promoter region) recognised by the chosen polymerase
- **NTPs** (nucleoside triphosphates)
- **Magnesium**, a co-factor for RNA polymerisation
- **Polycations** (spermine, spermidine), increase transcription efficiency

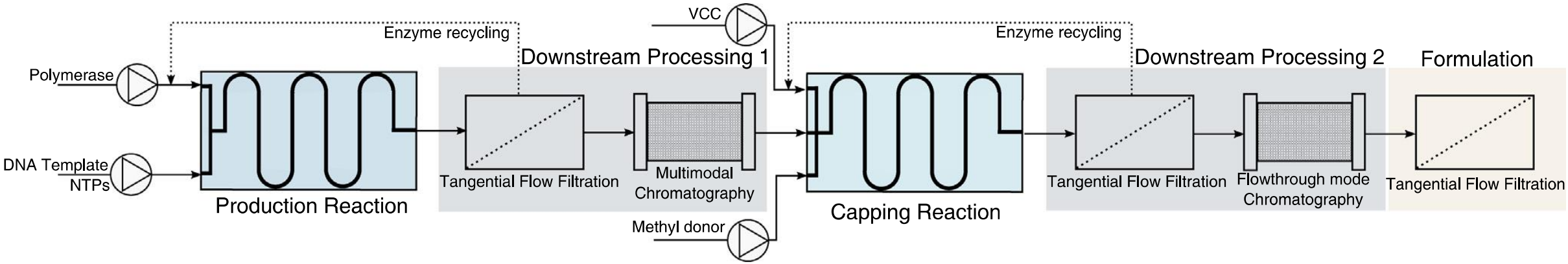
Example: production of mRNA for vaccines

- Addition of the 5' cap is typically **done enzymatically after the initial transcription reaction**
- The poly-A tail can be **encoded in the DNA template, or added enzymatically** at a later stage using a non-template dependent RNA polymerase (poly-A polymerase)



Example: production of mRNA for vaccines

- Below a **conceptual design** for mRNA manufacturing (*Vaccine* 39 (2021) 2190–2200)
- **Two production steps**, IVT and enzymatic capping
- Purification via **ultrafiltration** allows for enzyme recycling
- Additional purifications with **chromatography** (ion-exchange, size exclusion, affinity using poly-A tag…)

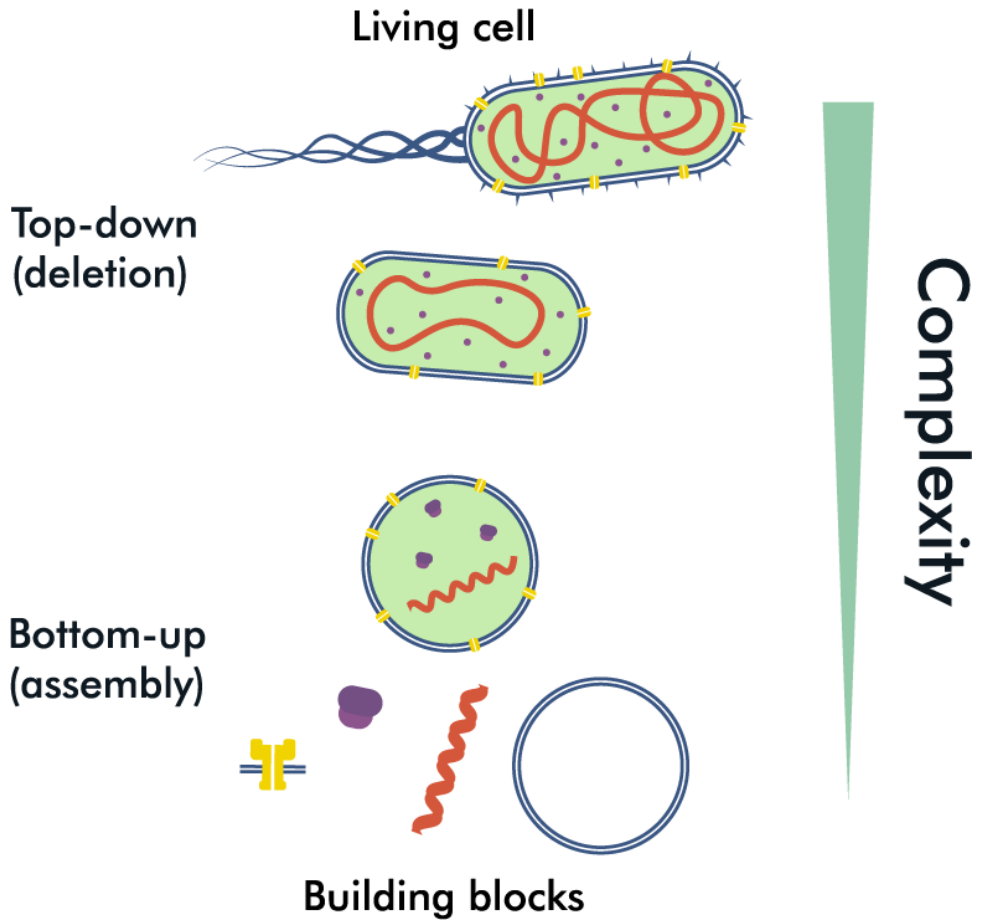


Lecture outline

1) Cell free systems and bioprocessing
   - Example: mRNA production
2) **Top-down synthetic biology: Minimal cells**
3) Bottom-up synthetic biology: Synthetic cells

Top-down and bottom-up synthetic biology

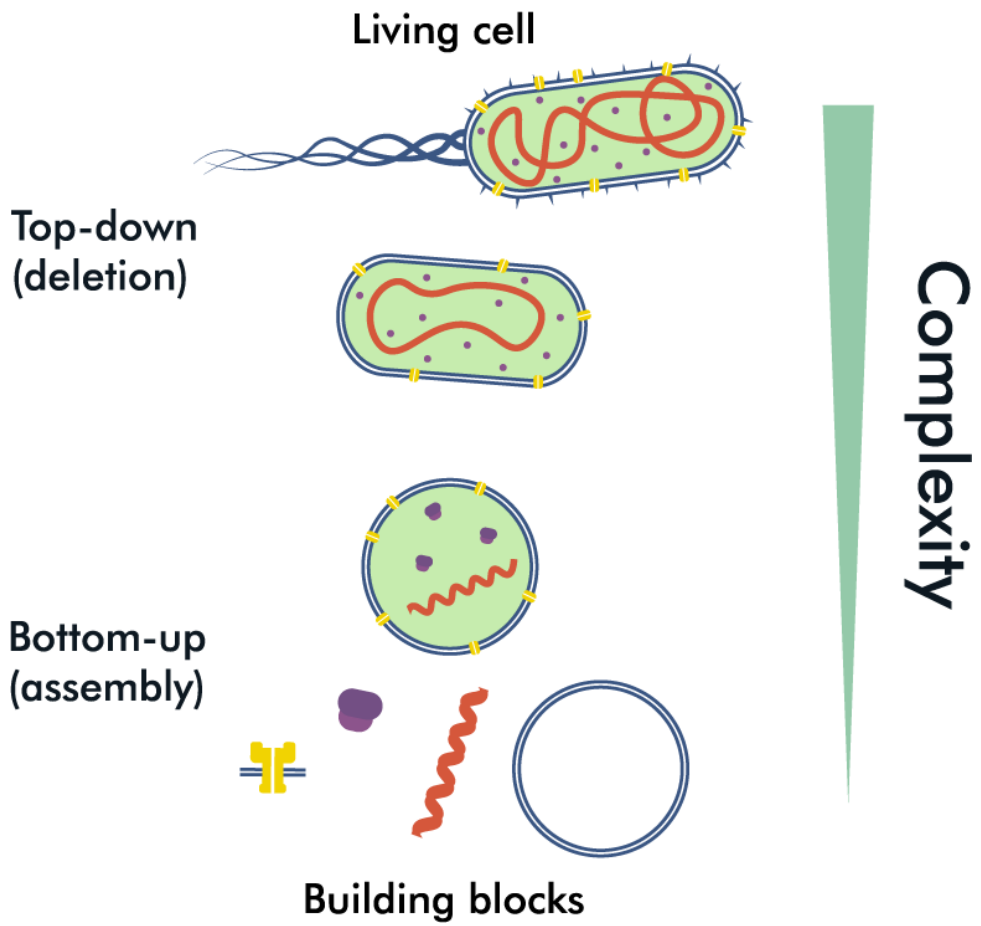
- **Synthetic biology** aims to apply quantitative, **engineering principles** to modify living systems and/or create new biological parts of systems
- This can be done in two ways, following a "**top-down**" or a "**bottom-up**" approach



QRB Discovery, 6: e5, 1–9
https://doi.org/10.1017/qrd.2024.27.

Top-down and bottom-up synthetic biology

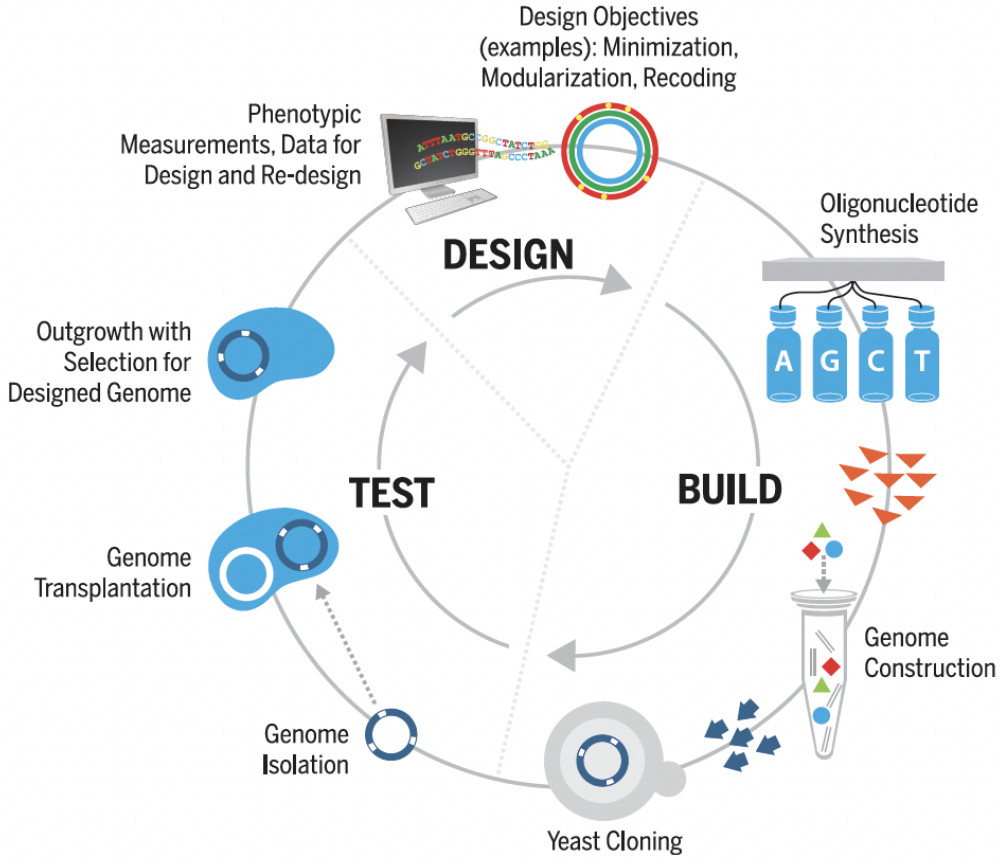
- In **top-down** Synthetic Biology one starts from a living cell (or organism), and attempts to modify it, **often reducing its complexity** or optimising it for a specific function, e.g. the production or transformation of a given compound
- Invariably this is done through **genetic engineering**, as we have studied extensively



QRB Discovery, 6: e5, 1–9
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Example: Miminal cells

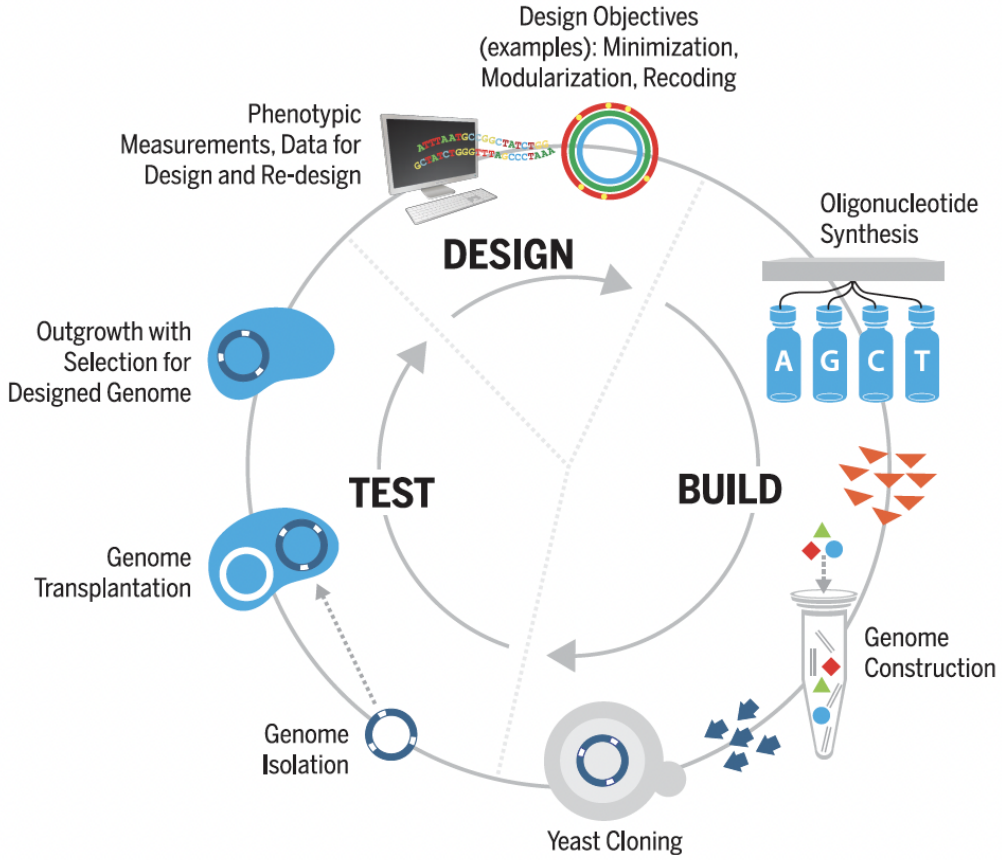
- Humans have ~20000 genes
- "Simple" bacterial cells like *E. coli* and *B. subtilis* have ~4000 to ~5000 genes.
- Many of these genes, however, are **not strictly required to sustain life;** they are needed so the cells can **adapt** to the very different growth conditions it encounters



C. A. Hutchison III et al. Science, 351, 6280, 2016

Miminal cells

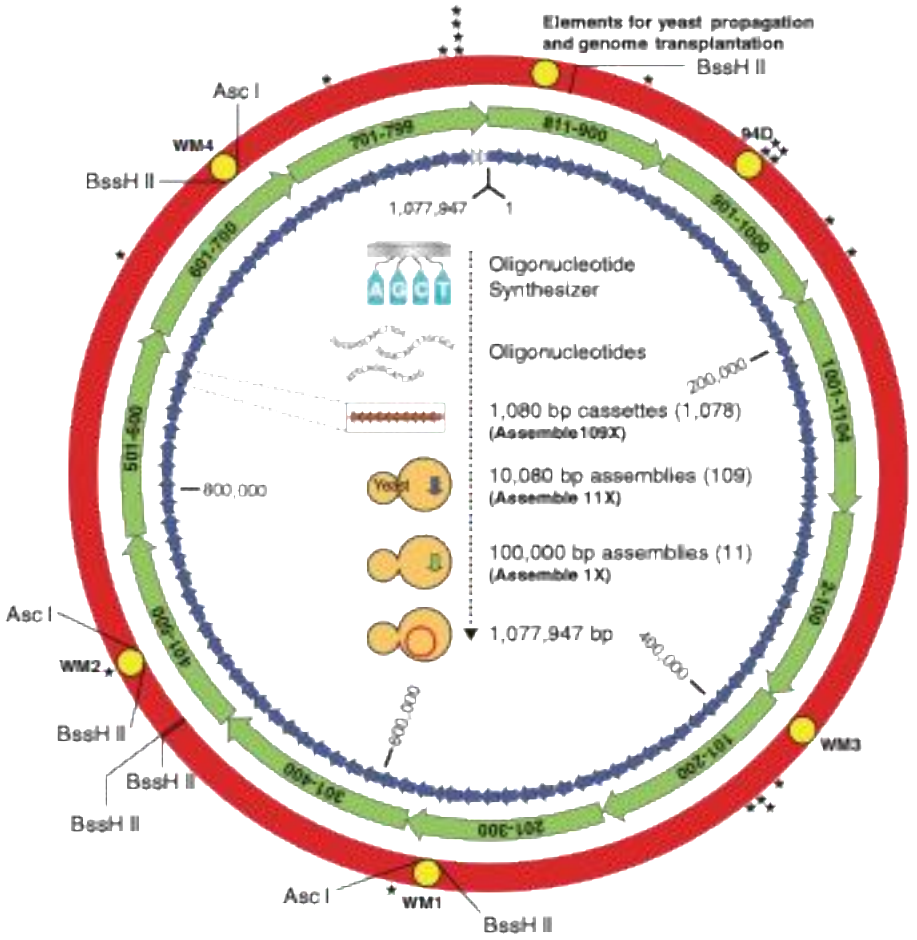
- Cells that live in stable environments have smaller genomes. Non-essential genes needed for adaptation are lost.
- *Mycoplasma* are a class of simple bacteria that tend to live in mammalian hosts
- These are stable environments, so no need for adaptation
- *Mycoplasma mycoides* (Lecture 2) has ~1000 genes (divides in 60 minutes). *Mycoplasma genitalium* only has 525 genes (but divides in several hours)!



C. A. Hutchison III et al. Science, 351, 6280, 2016

Miminal cells

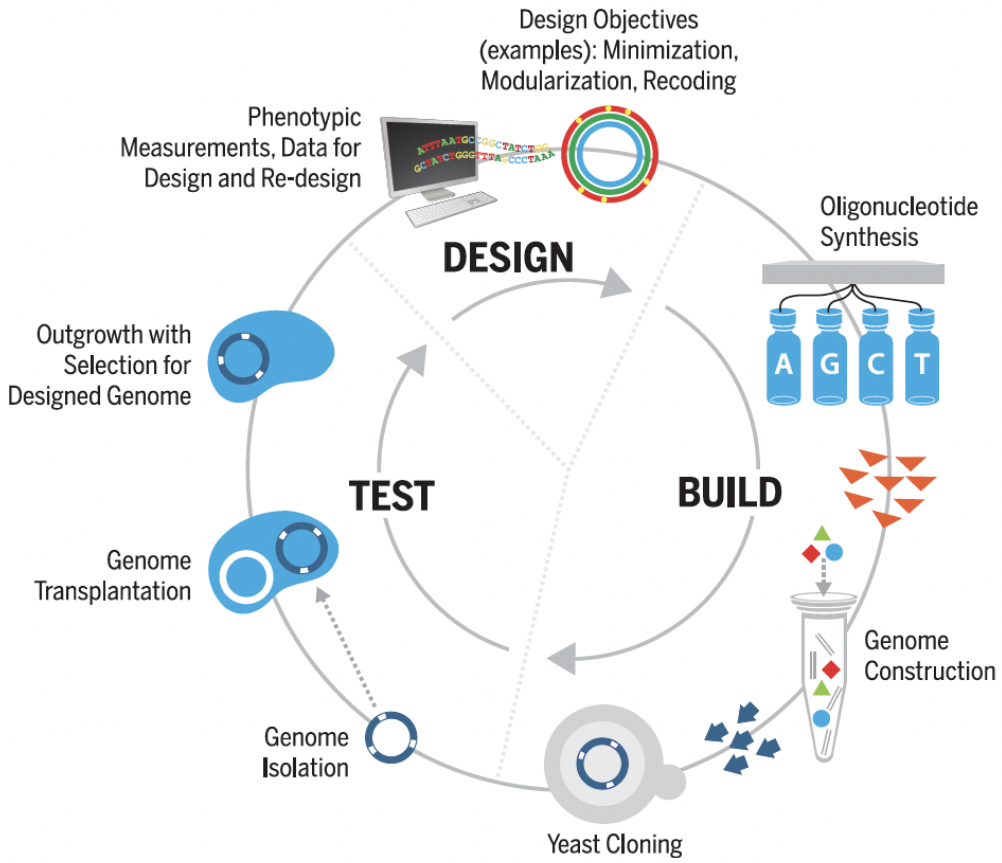
- From Lecture 2: researchers from the Craig Venter Institute (JCVI) crated a fully synthetic genome of *M. mycoides*: **JCVI syn1.0**
- The same team attempted to create a **minimal genome**, that contains only the genes that are essential to sustain life
- **Why?** Building a minimal cell is very interesting: we can learn about the **essential functions for life**



Gibson et al, Science 329, 53-56, 2010

Miminal cells

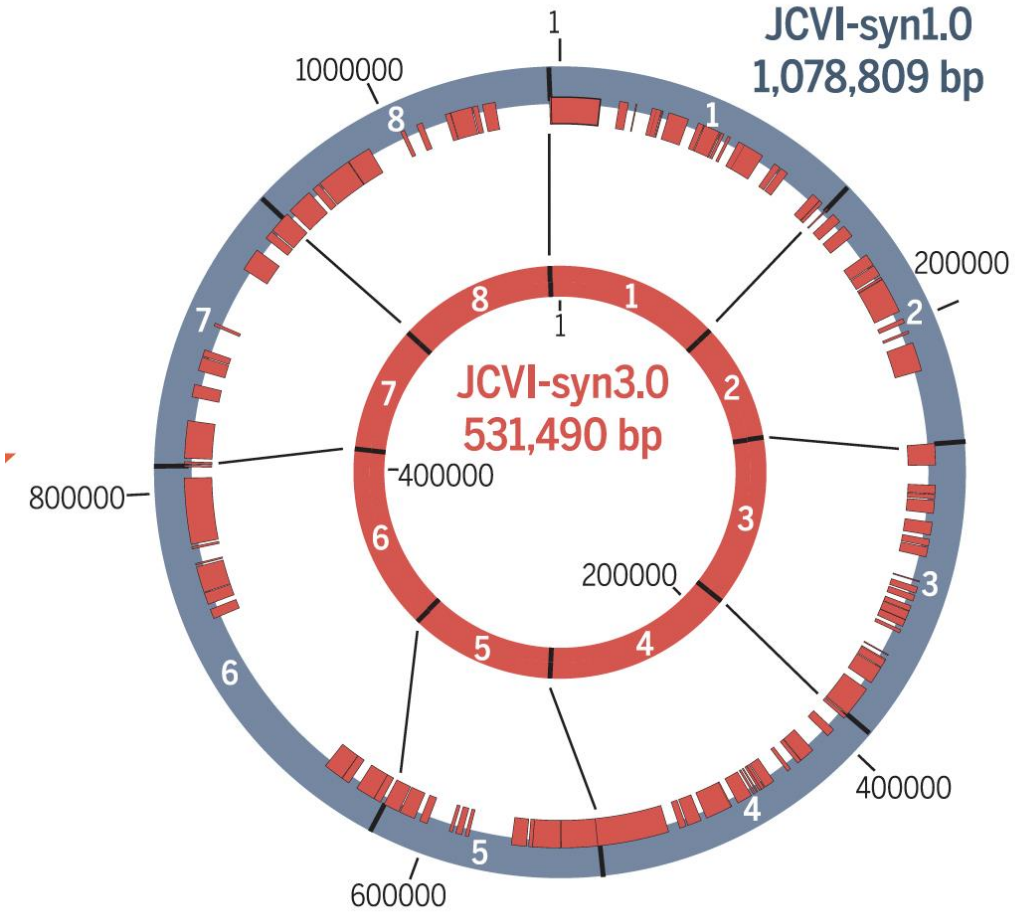
- Started from **JCVI syn1.0** rather than *M. genitalium*, as the latter divides too slowly for practical use
- Used mutation analysis and bioinformatics (and iterative optimisation) to identify essential genes
- Ultimately, created **JCVI syn3.0**, defined as "**a working approximation of a minimal cellular genome**"



C. A. Hutchison III et al. Science, 351, 6280, 2016

Miminal cells

- JCVI syn3.0 has **473 genes**
- 438 genes code for proteins, 35 for RNAs
- Contains essential genes and "quasi essential" genes
- Removing "quasi essential" genes leads to very slow growth
- Of the 473 genes, **149 have unknown biological function**

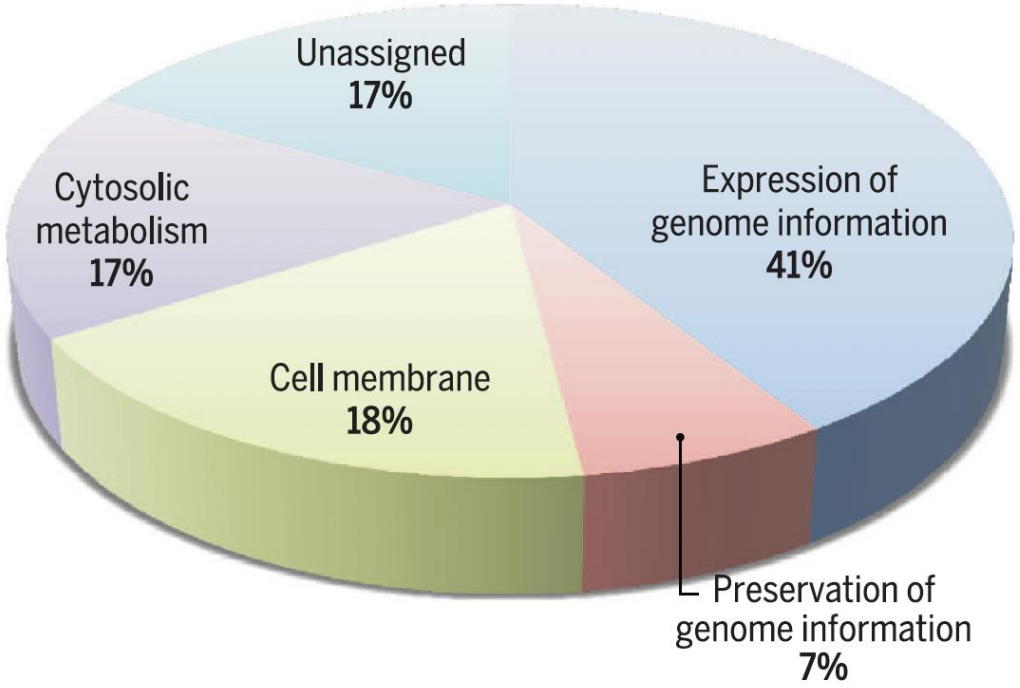


C. A. Hutchison III et al. Science, 351, 6280, 2016

Miminal cells

Most of the retained genes can be classified into four functional categories

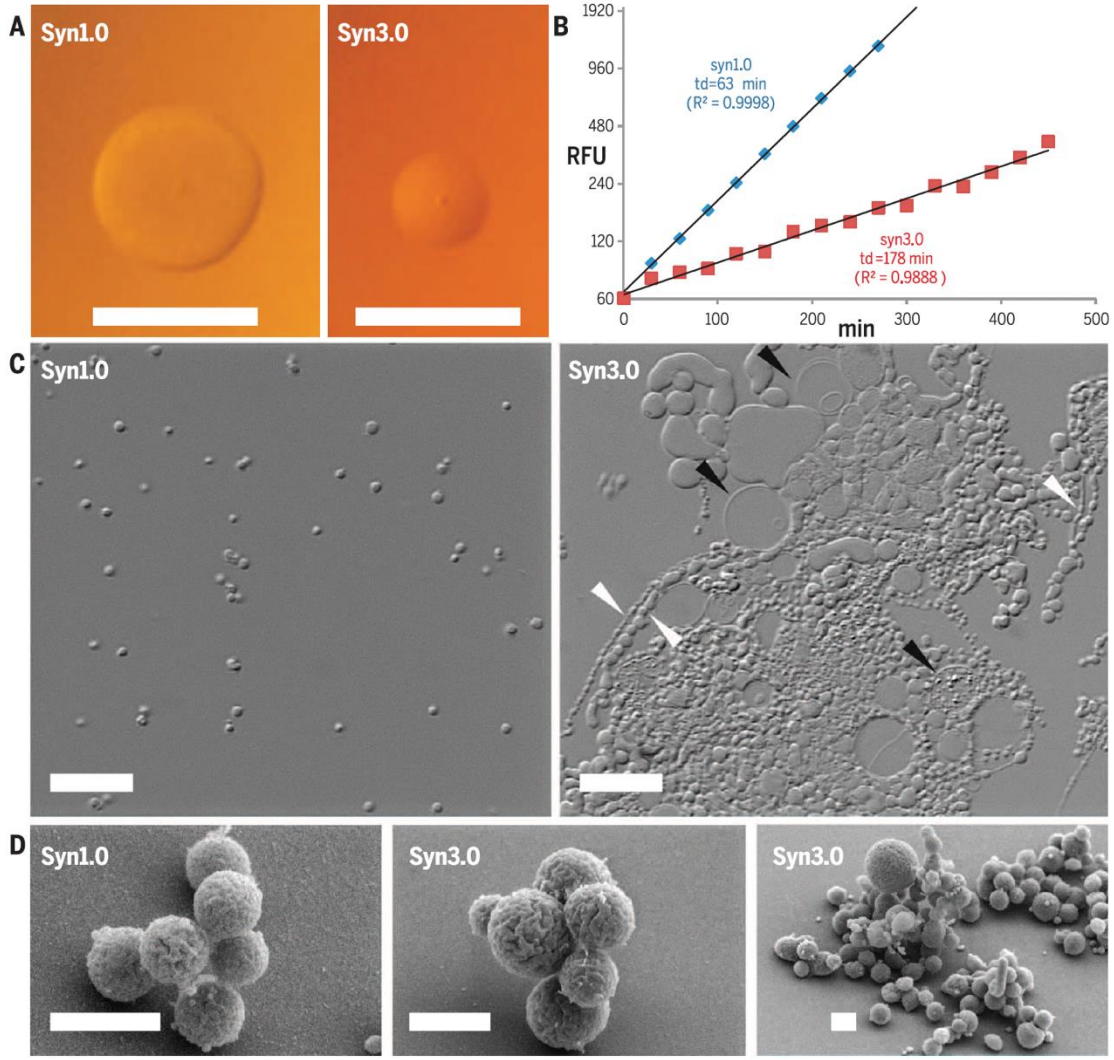
1. Expression of genome information
2. Preservation of genome information
3. Cytosolic metabolism
4. Cell membrane (interesting how many genes are needed for this!)



C. A. Hutchison III et al. Science, 351, 6280, 2016

Miminal cells

- JCVI syn3.0 grows slowly, ~3h per vision cycles, but still much faster than *M. genitalium*
- Cells have irregular morphologies and are polydisperse



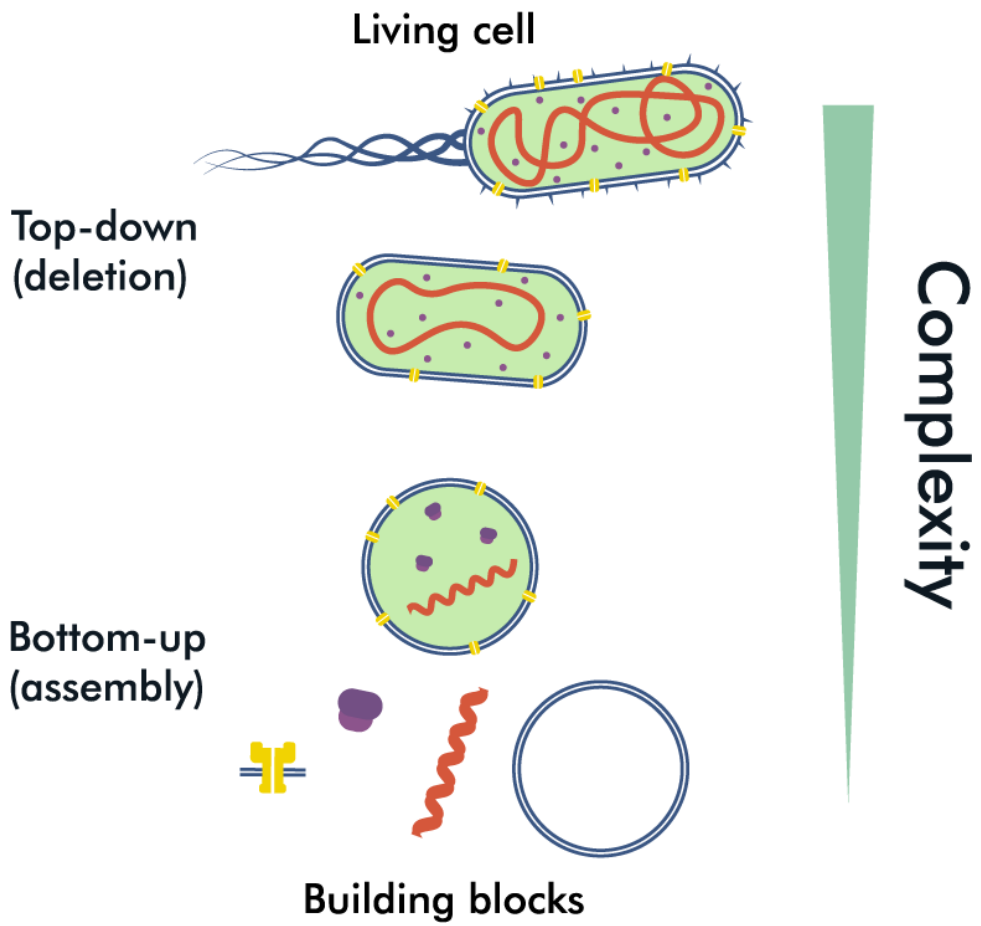
C. A. Hutchison III et al. Science, 351, 6280, 2016

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   - Example: mRNA production
2) Top-down synthetic biology: Minimal cells
3) **Bottom-up synthetic biology: Synthetic cells**

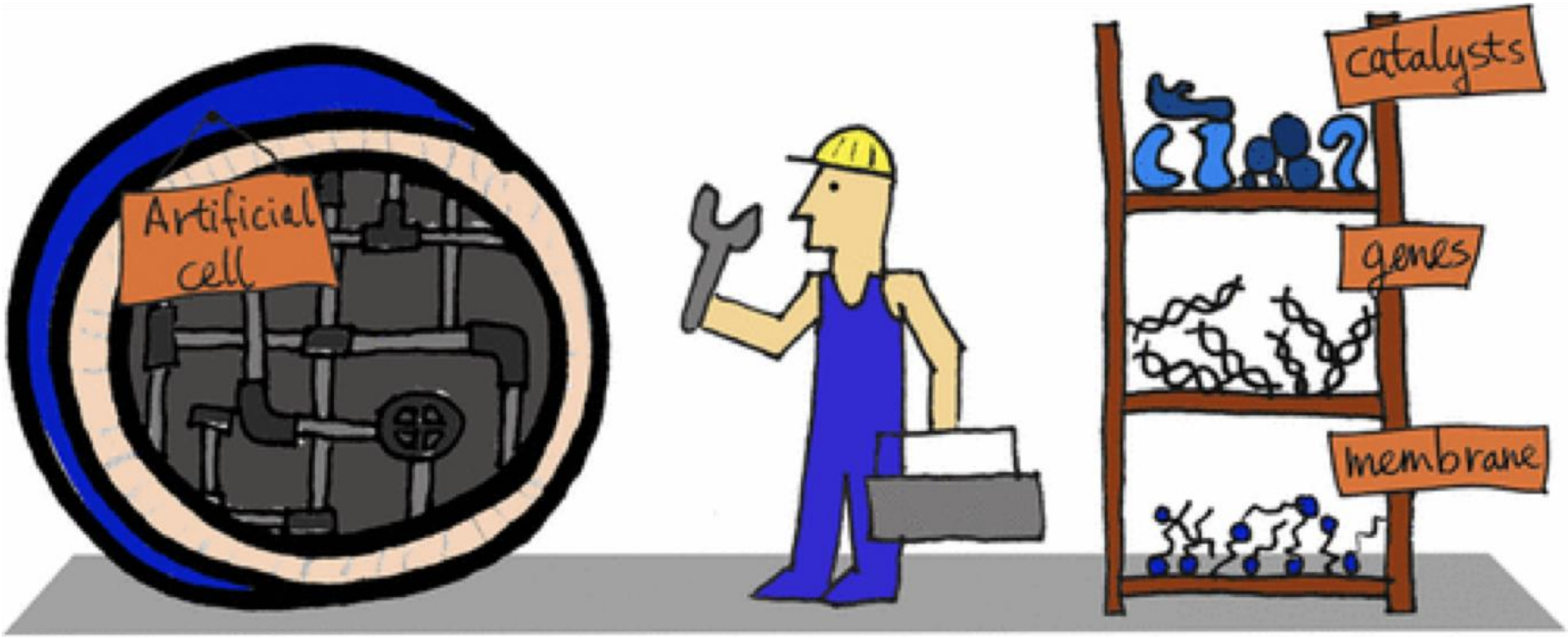
Bottom-up synthetic biology

- Building a minima cell **is still top-down**. We start from a natural genome (doesn't matter it is made synthetically!) and remove genes
- In **bottom-up synthetic biology** we aim to start from **non-living**, **elementary components** (building blocks) and combine them to create fully synthetic biological or biomimetic systems



QRB Discovery, 6: e5, 1–9
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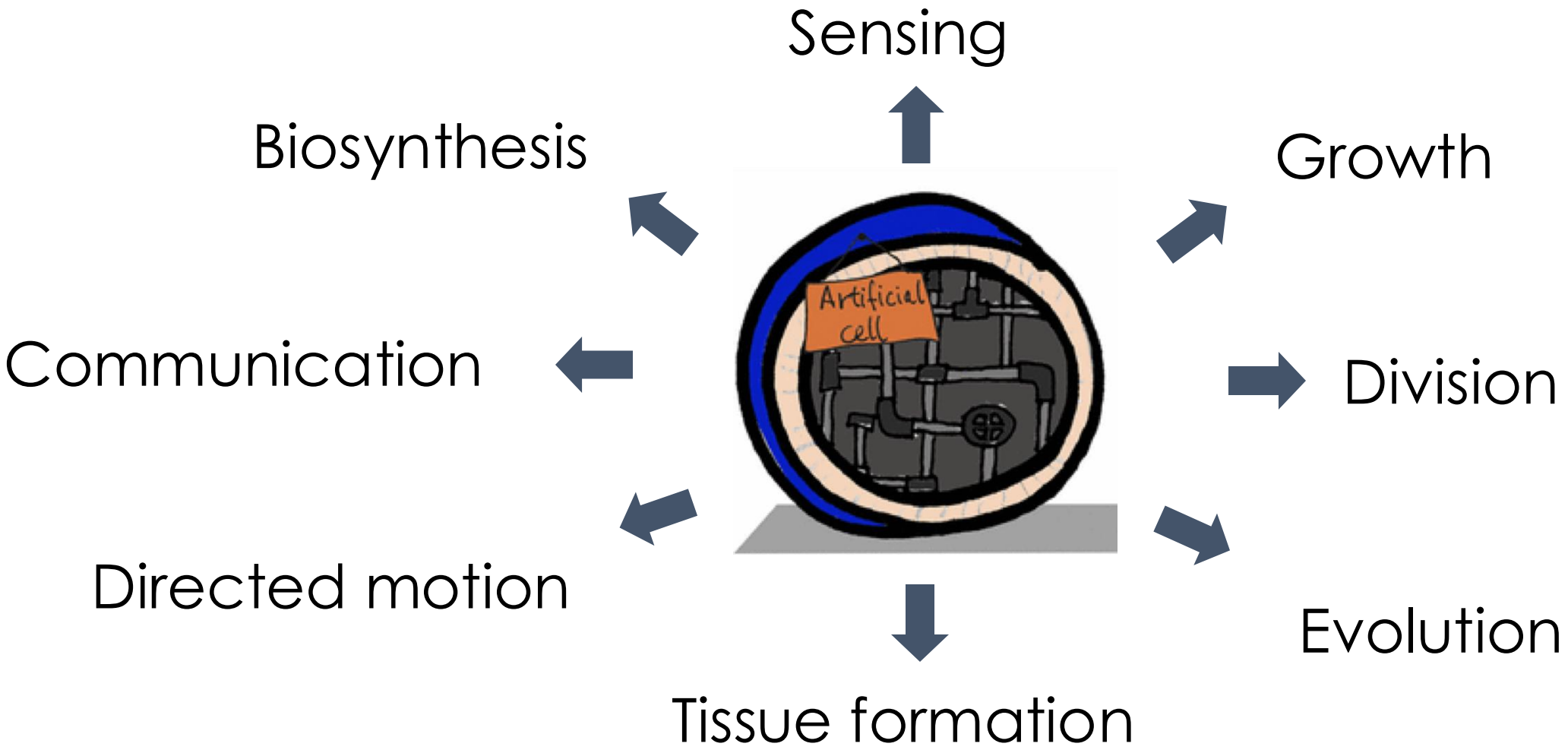
Synthetic Cells (SynCells)



Acc. Chem. Res., 50 769–777, 2017

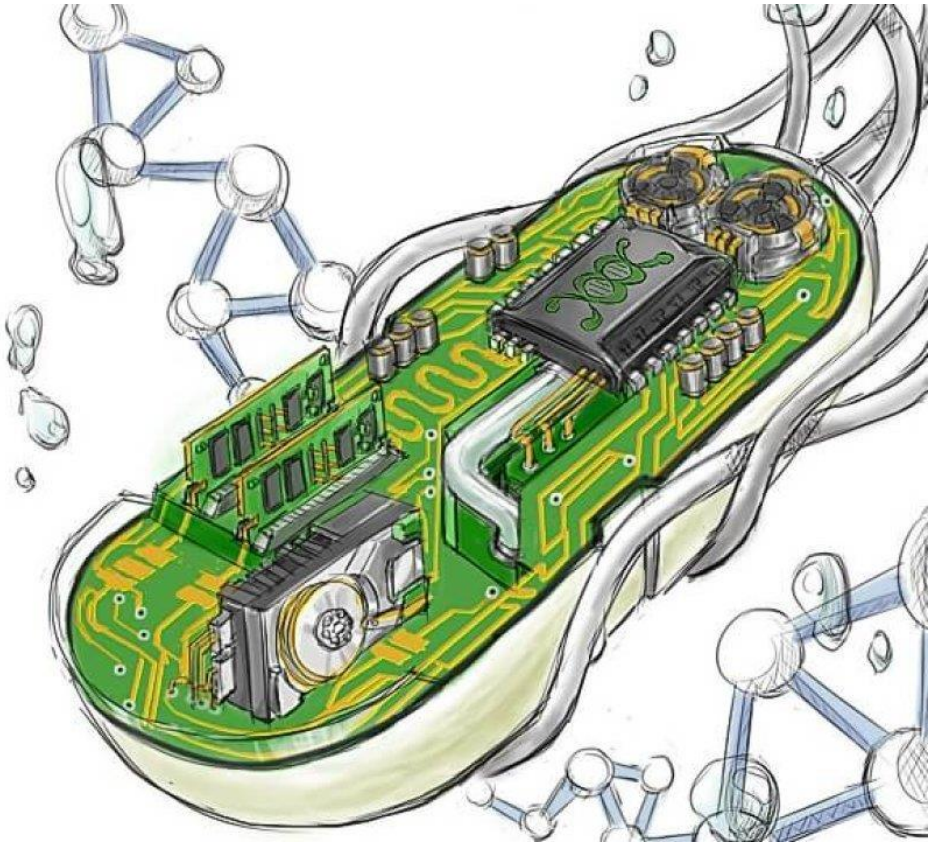
Cell-like "microrobot" or "microreactor" constructed from the bottom-up to imitate cellular structure and function

SynCell Functions



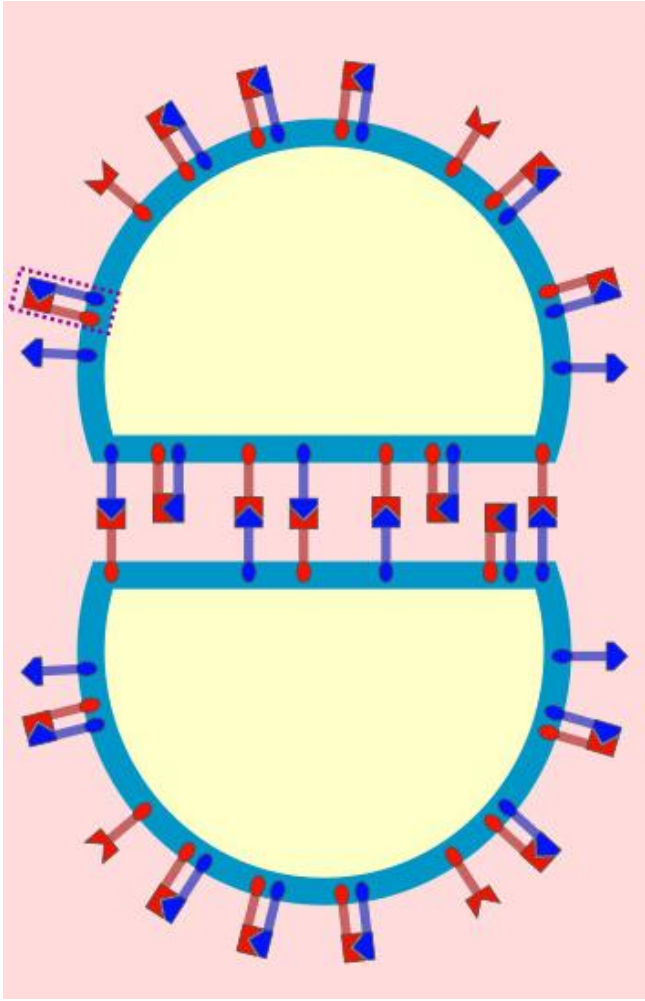
SynCell Applications: Learning about biology by building new biology

- Sometimes it is difficult to study biological mechanisms in cells, as they are too complicated
- Some can be reconstituted in vitro, but it may not be sufficiently realistic
- Synthetic cells offer an in-between approach: one can replicate a specific biological process in isolation, but in a more realistic environment



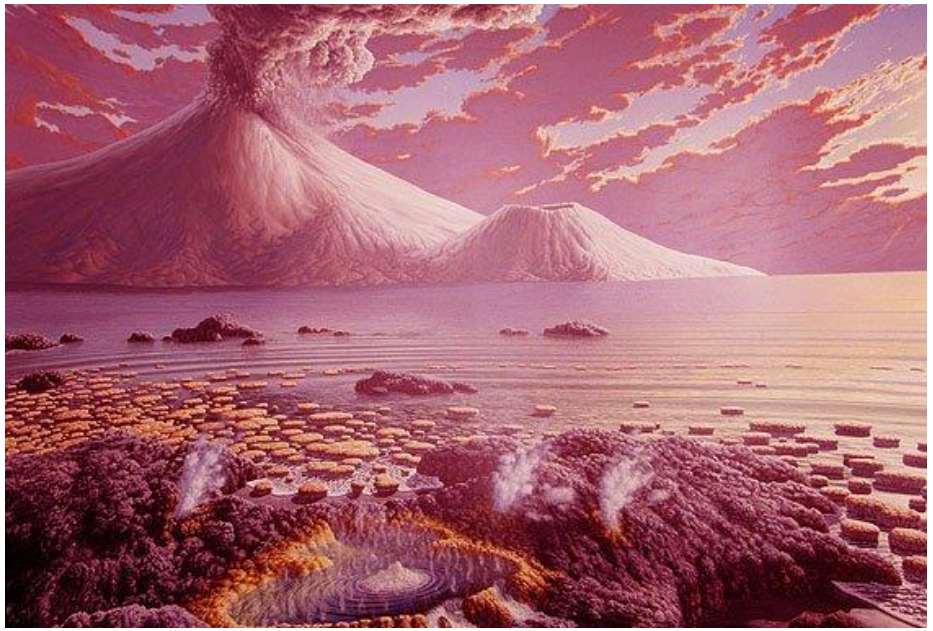
SynCell Applications: Learning about biology by building new biology

- Examples:
  - Gene expression in confinement (why is this interesting?)
  - Trans-membrane signalling
  - Communication and population dynamics
  - Cell-membrane biophysics (membrane adhesion, fusion, splitting)



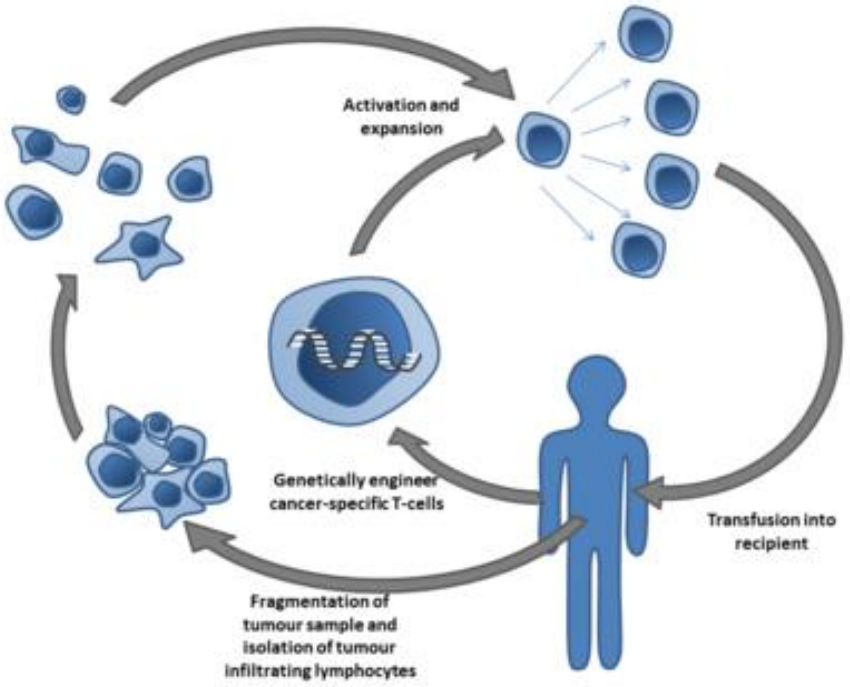
SynCell Applications: Origin of life

- Cell are the basic units of life, but we have no idea of what the first cell looked like, or how it formed
- Intense research into the origin of life, often aiming to build **prebiotically-relevant protocells**
- These should form from simple molecules that may have existed before complex building blocks emerged (RNA, fatty acids…)



SynCell Applications: Technological applications

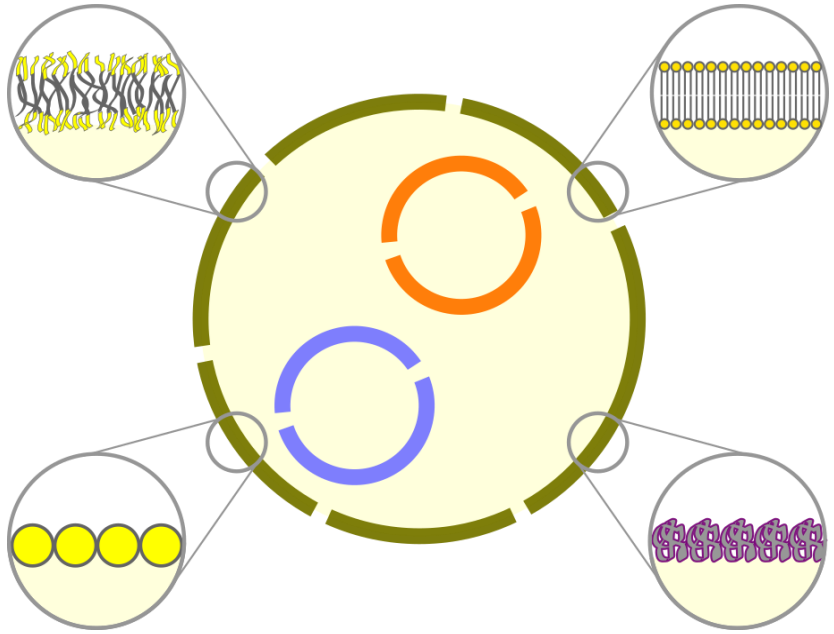
- **Synthetic cell therapies**: Can we build a synthetic cell that acts like an immune cell and use if for cellular therapies? This would potentially be cheaper and safer than cell-based therapies
- **Smart vaccines and drug delivery vectors**: using complex architectures with multiple compartments, sustaining active processes, targeting disease sites
- **Biomanufacturing**
- **Environmental remediation and resource recovery**



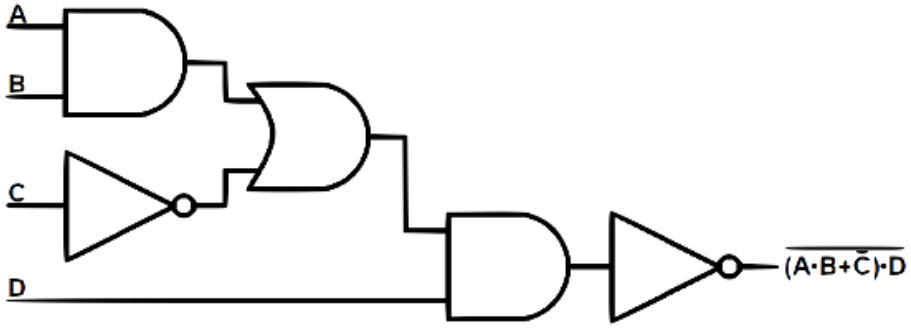
How to build a SynCell

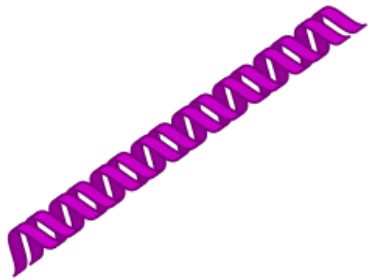
Structure

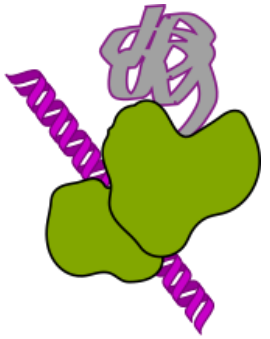
Micro-compartmentalized architectures



Information Processing

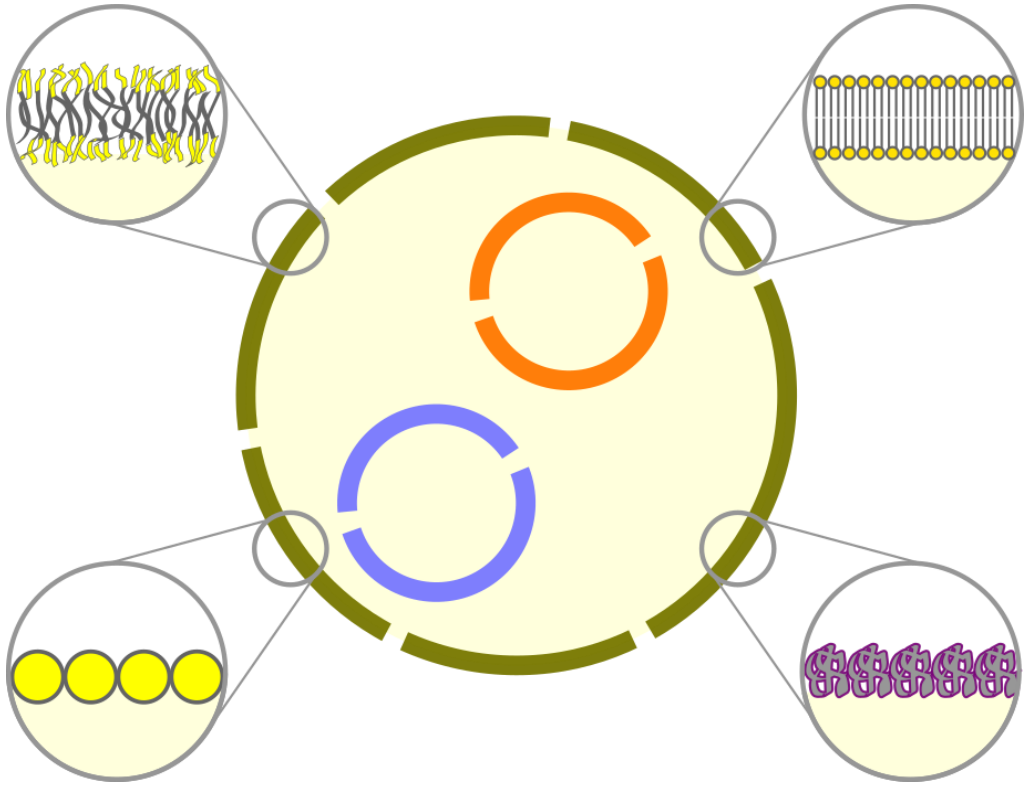






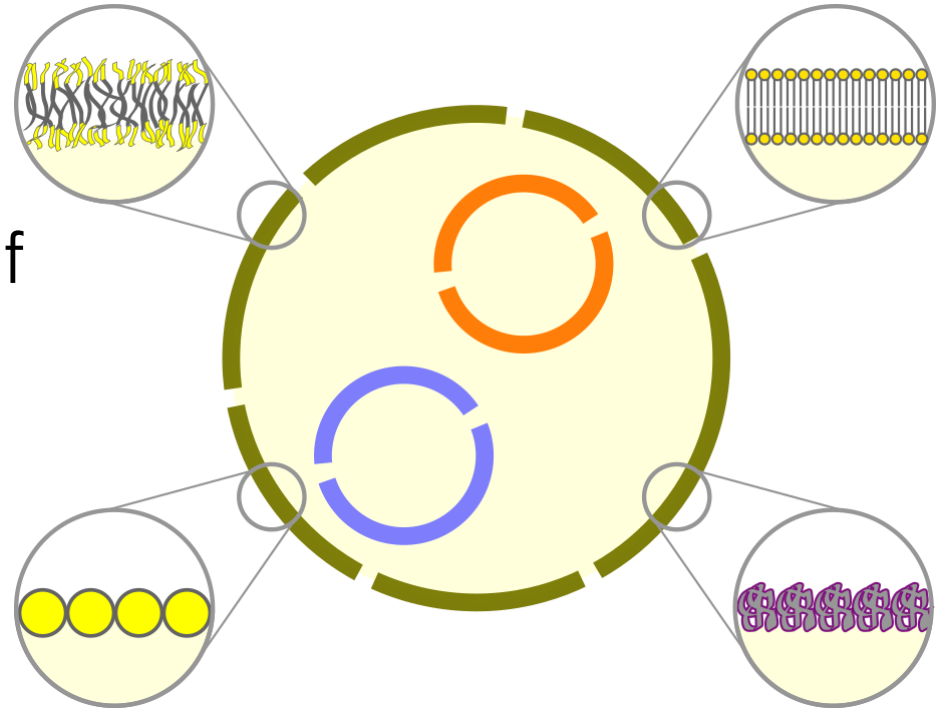
SynCell Compartments: Membranes

- **Compartmentalisation** is one of the most fundamental features of cells
- **SynCells need compartments**, analogous to the plasma membrane, that separate the inside from the outside of the SynCell
- These often (but not always) comprise synthetic membranes, which can be made from
  - **Phospholipids**
  - **Synthetic block copolymers**
  - Cross-linked proteins
  - Nanoparticles/colloidal particles



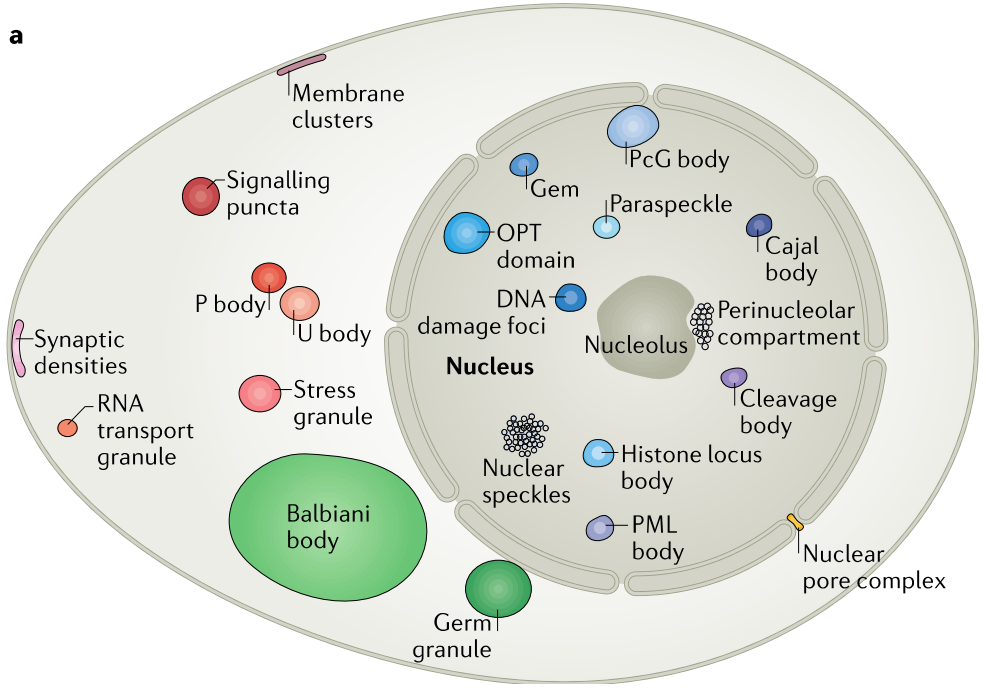
SynCell Compartments: Membranes

- **Different materials offer advantages/disadvantages**. E.g. lipid membranes have the advantage of being the most similar to cell membranes, but polymer membranes are more robust
- **Lipid vesicles** (spherical membranes) are the most common choice. We can make them of different sizes:
  - Giant Unilamellar Lipid Vesicles (GUVs), ~1-100 microns
  - Large Unilamellar Lipid Vesicles (LUVs), 0.1-1 microns
  - Small Unilamellar Lipid Vesicles (SUVs), <0.1 microns
- **Internal compartments** may be required to spatially organise biochemistry: **SynCells may need synthetic organelles**



SynCell Compartment: Membrane-less

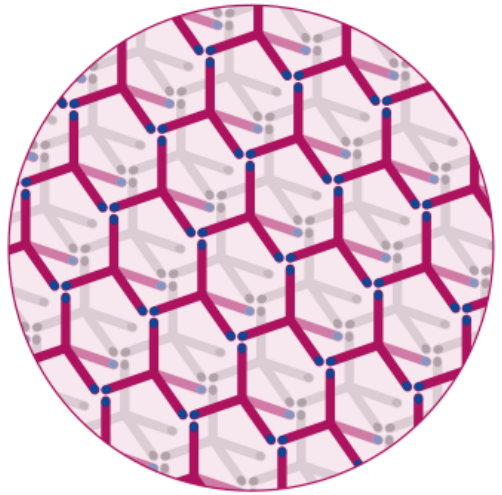
- Not all compartments are membrane-bound!
- Cells contain droplets enriched in proteins and nucleic acid that form in the cytoplasm or nucleus due to Liquid-Liquid Phase Separation: **Biomolecular Condensates**
- Condensates can sequester or exclude clients (enzymes, substrates, nucleic acids) based on affinity for the dense phase. The are **membrane-less compartments**.
- Cells use condensates to control biochemical reactions and regulate homeostasis. They are functional and referred to as **membrane-less organelles**

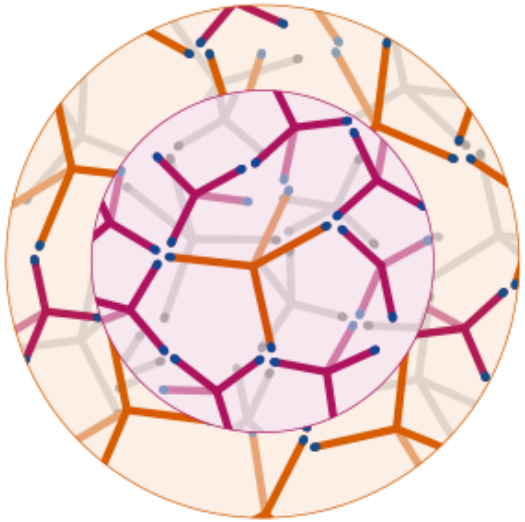


Nat. Rev. Mol. Cell Biol., 18, 285–298 (2017)

SynCell Compartment: Membrane-less

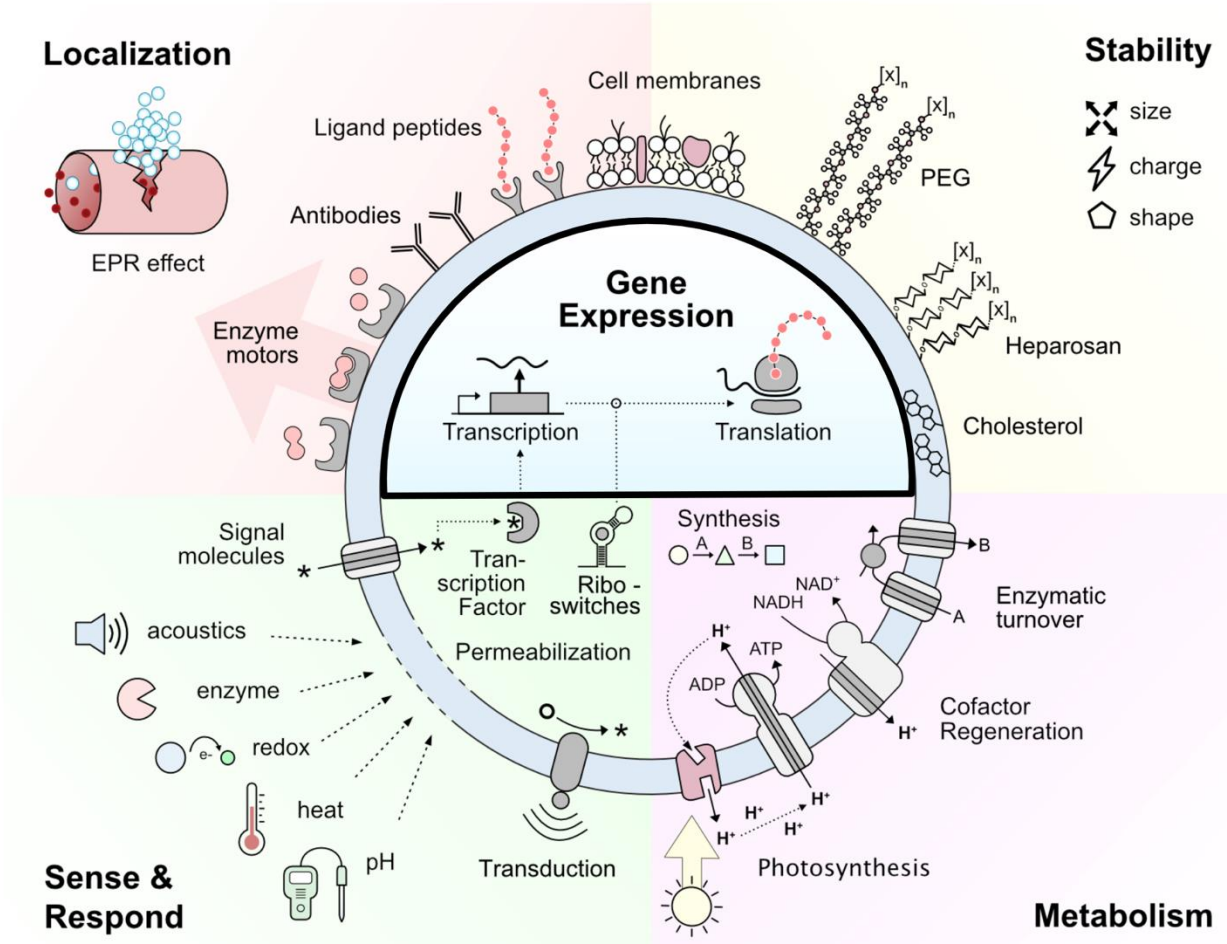
- Membrane-less compartments can be used to create SynCells and synthetic organelles. These can be prepared from
  - Synthetic proteins/peptides
  - Charged polymers (complex coacervates)
  - Cross-linked polymers (hydrogels)
  - DNA or RNA nanostructures
- Membrane-less compartment are **more robust and dynamic** compared to membranes, but they don't stop diffusion of products (it's harder to stablish strong gradients)





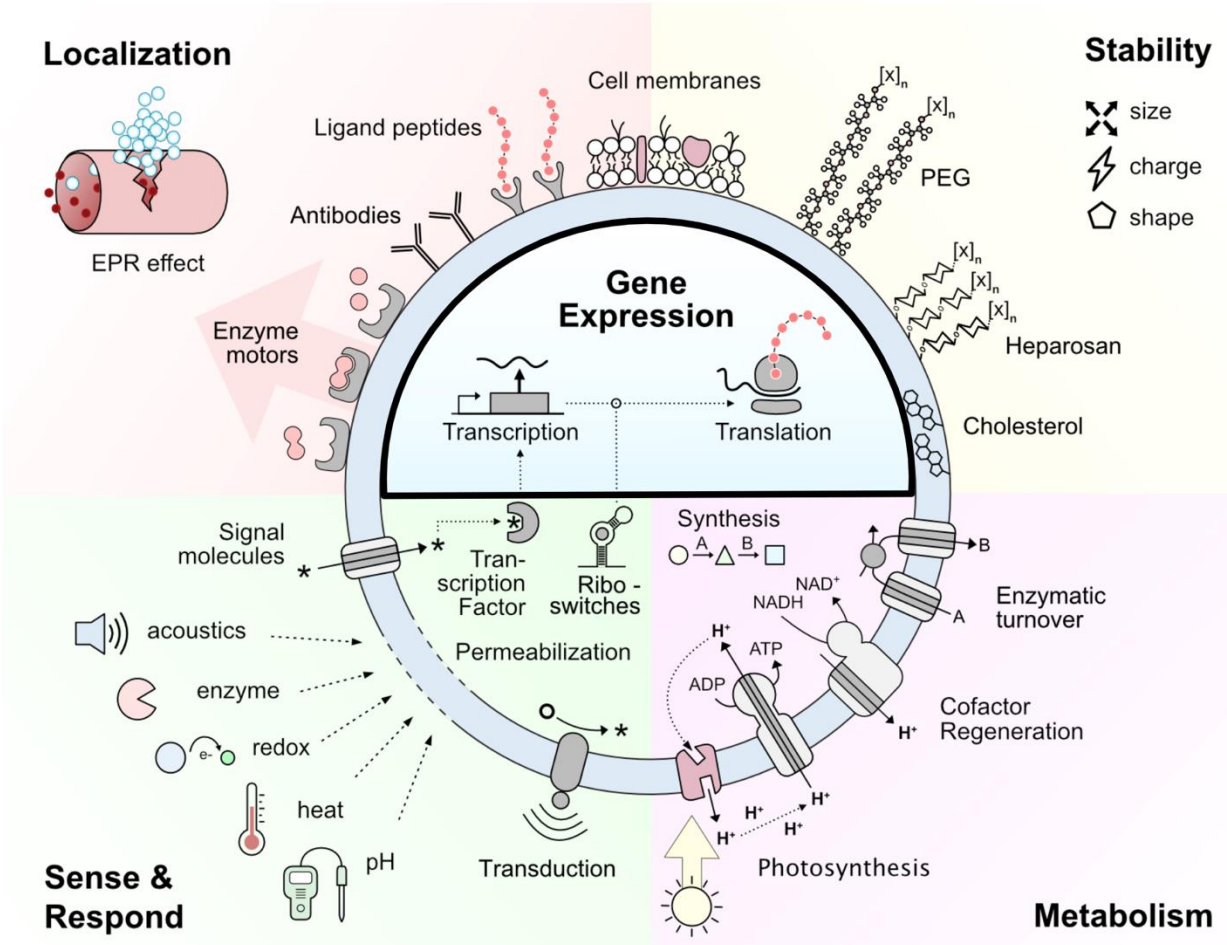
SynCell Machinery

- To sustain life-like functions, SynCells need **programmable molecular machinery** that operates onto or within the compartments. Options include
- **Biological machinery**
  - Transcription/translation machinery
  - Enzymes
  - Membrane proteins (pores, ligands, receptors)
  - Nucleic acids (genes, RNAs)



SynCell Machinery

- **Non-biological machinery**
  - Nanomaterials (e.g. nanoparticles)
  - Synthetic polymers
  - Synthetic nucleic acid nanostructures
- The possibility of **combining biological and non-biological materials** is one of the strengths of SynCell engineering



SynCells expressing synthetic organelles…

